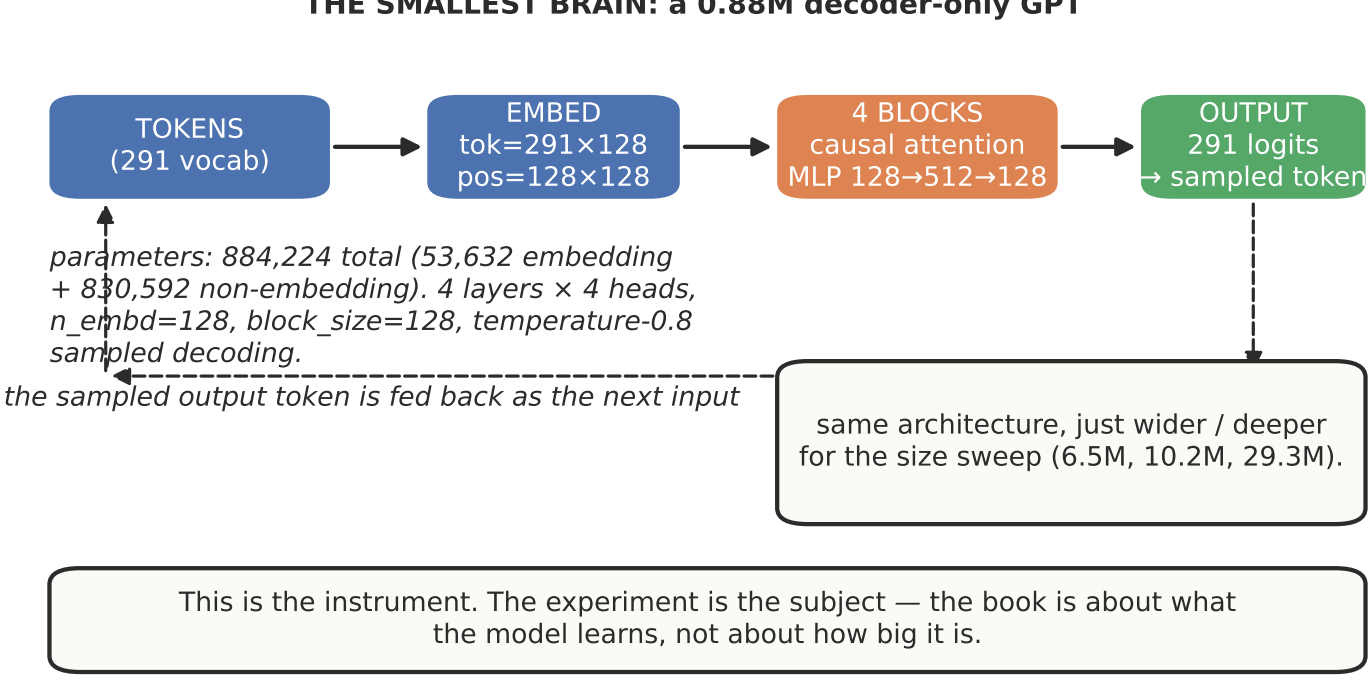
THE SMALLEST BRAIN: a 0.88M decoder-only GPT
TOKENS
(291 vocab)
EMBED
tok=291×128
pos=128×128
4 BLOCKS
causal attention
MLP 128→512→128
OUTPUT
291 logits
→ sampled token
parameters: 884,224 total (53,632 embedding
+ 830,592 non-embedding). 4 layers × 4 heads,
n_embd=128, block_size=128, temperature-0.8
sampled decoding.
the sampled output token is fed back as the next input
same architecture, just wider / deeper
for the size sweep (6.5M, 10.2M, 29.3M).
This is the instrument. The experiment is the subject — the book is about what
the model learns, not about how big it is.